

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
6	(a)	(i)	Waves from source interfere (accept superpose) with waves <u>reflected</u> [from plate] (1) Waves in phase or constructively interfere at [each] antinode (1)		2		2		
		(ii)	Antinodes $\frac{\lambda}{2}$ apart or $\lambda = 32 \text{ mm}$ (1) 8 mm = $\frac{1}{4} \lambda$ or equivalent, so node at P (1) If correct answer and valid argument without any reference to λ award 1 mark. Alternative for 1 mark only – phase change of 180° <u>on reflection</u> therefore node at P		1	1	2		
	(b)	(i)	In antiphase. Accept exactly, or π , or <u>180° out of phase</u> or half a cycle (1) don't accept phases expressed in terms of λ Longest λ is 0.60 m [because $0.30 \text{ m} = \frac{\lambda}{2}$] (1) Second longest is 0.20 m [because $0.30 \text{ m} = \frac{3\lambda}{2}$] (1)	1	1 1		3		
		(ii)	$f = 20 \text{ [Hz]}$ or $T = 0.050 \text{ s}$ if <u>used</u> in $v = \frac{\lambda}{T}$ (1) If $\lambda = 0.60 \text{ m}$, then $v = 20 \times 0.6 = 12 \text{ [m s}^{-1}\text{]}$ (1) ecf on both λ This is in range; $\lambda = 0.20 \text{ [m]}$ is out of range (1) Alternative method for last two marks 10 m s^{-1} at 20 Hz corresponds to $\lambda = 0.50 \text{ m}$, and 15 m s^{-1} at 20 Hz corresponds to $\lambda = 0.75 \text{ m}$ (1) So $\lambda = 0.60 \text{ m}$ (1) ecf on (i) extending to conclusion (also rewards strategy)		3		3	2	
			Question 6 total	1	8	1	10	2	0